

School Website with Student Portal Implementation Plan and Procedure

Project Title

School Information and Student Portal System Using GitHub Pages and Firebase

1. Project Objectives

General Objective: Develop a secure, cloud-based school website with an integrated student portal.
Specific Objectives: Create a public school website. Implement secure user authentication. Develop student, teacher, and admin portals. Manage grades, attendance, and announcements. Deploy the system online at zero hosting cost. Ensure data privacy through role-based access control. **2.**

Scope of Work

Public Website: Home, About, Faculty Directory, Announcements, Contact Information, Login Portal.

Student Portal: Login, Dashboard, Grades, Attendance, Announcements, Profile Management.

Teacher Portal: Login, Grade Encoding, Attendance Recording, Announcement Posting.

Admin Portal: Student, Teacher, Subject, Announcement, and User Management.

3. Technology Stack

Frontend: HTML5, CSS3, JavaScript, Bootstrap 5

Hosting: GitHub Pages

Authentication: Firebase Authentication

Database: Firebase Firestore

Storage: Firebase Storage

Version Control: GitHub

4. Project Phases

1. Planning and Analysis (1 Week)
2. UI/UX Design (1 Week)
3. Firebase Configuration (2–3 Days)
4. Public Website Development (1 Week)
5. Authentication Module (3 Days)
6. Student Portal Development (1 Week)
7. Teacher Portal Development (1 Week)
8. Admin Portal Development (1 Week)
9. Security Implementation (3 Days)
10. Testing (1 Week)
11. Deployment (1 Day)

5. Firebase Collections

users, students, teachers, subjects, grades, attendance, announcements

6. Security Model

Students: Access own records only.

Teachers: Access assigned class records only.

Administrators: Full access.

7. Testing

Functional Testing, Security Testing, User Acceptance Testing.

8. Proposed Timeline

Total Estimated Duration: 8–10 Weeks.

9. Success Criteria

Website accessible online, secure authentication, role-based access control, mobile compatibility, and zero hosting cost within free-tier limits.